GRASP: Greenwashing Risk Assessment in Structured Business Responsibility and Sustainability Reporting in India

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Abstract

India's Business Responsibility and Sustainability Reporting (BRSR) framework produces a large collection of corporate disclosures that combine narrative claims with numerical information. Reviewing these reports requires analysts to locate supporting evidence and compare disclosures across companies. GRASP supports this review through a seven-module pipeline. Reports are collected and stored with company, period, and measurement details. Report tags and concept mappings automatically assign principle labels to selected text segments, creating a weakly labelled dataset. A BERT language model adapted to BRSR text is then trained to assign each text segment to one of nine reporting principles. Word lists and sentence similarity identify vague wording. Numerical and assurance checks create review records linked to the reported evidence. Classifier probabilities provide principle-level summaries, and a Transparency Score combines Completeness, Specificity, Clarity, and Sustainable Development Goal (SDG) breadth. The vagueness detector is reused to calculate Clarity. Company and sector comparisons are made available through an application programming interface. The application outputs cover 1,164 companies. On 6,201 text segments from 175 companies excluded from supervised training, the classifier achieves 93.00% agreement with report-derived weak labels. The Transparency Score has a Pearson correlation of $r = 0.6314$ with 20 historical human ratings. By connecting summary scores with source disclosures, GRASP helps analysts identify review priorities and inspect the evidence behind each result.

Keywords: BRSR; sustainability disclosure; weak supervision; BERT; vagueness detection; transparency scoring; sector benchmarking.

1 Introduction

India's Business Responsibility and Sustainability Reporting (BRSR) framework provides a common structure for environmental, social, and governance (ESG) disclosures. The Securities and Exchange Board of India (SEBI) introduced mandatory BRSR reporting for the top 1,000 listed companies by market capitalisation from financial year 2022–23. The framework covers nine principles of responsible business conduct, with mandatory essential indicators and voluntary leadership indicators [1]. Each annual report combines written explanations with numerical information. In this paper, a *text segment* is a selected narrative response, and a *fact* is an individual reported value stored with the relevant reporting context.

Manual review requires more than reading the narrative. An analyst must identify the reporting principle, locate a related figure, and check the measure, unit, period, and business boundary. A claim such as *water use decreased* becomes meaningful only after a comparable water measure has been identified for both periods. Company comparisons add another requirement: differences in business activity and applicable reporting items must be considered. Repeating these checks across hundreds of reports makes manual analysis difficult to sustain at scale.

Existing methods address important parts of this workload. ClimateBERT analyses climate-related language, ChatReport supports source-linked questions about reports, and ESGReveal extracts structured information from ESG documents [2, 3, 4]. These approaches demonstrate the value of language models and retrieval for disclosure analysis. A connected BRSR workflow also needs a consistent link between claims and filed evidence, a documented method for obtaining training labels, and a transparent basis for company comparisons. Research on greenwashing detection and Indian disclosure practices supports treating language analysis, evidence review, and score interpretation as related but distinct tasks [5, 6].

GRASP addresses these requirements through a shared report database, a nine-principle classifier, evidence-linked review checks, and a four-component Transparency Score. Reporting tags and concept mappings provide *weak labels*: principle assignments generated from report structure rather than independent manual annotation. The resulting dataset contains 42,409 labelled text segments. Evaluation separates companies used for supervised training, validation, and testing, and also examines label sources and reporting concepts excluded from training. Application results show how the outputs support review at four levels, from a worked company example to comparisons across recorded sectors. The contribution is the integration of these functions in a traceable workflow, with the data group and reference source stated for each reported result.

The remaining sections follow the development and evaluation of this workflow. Section 2 reviews related research, and Section 3 presents the proposed solution. Section 4 explains the seven modules and score calculations. Sections 5 and 6 report model evaluations and company, sector, and year-wise results. Section 7 concludes the paper and outlines future work. Complete supporting records appear in the supplementary tables.

2 Literature Review

Research on automated sustainability analysis covers topic detection, commitment extraction, disclosure quality, evidence retrieval, and report-level decision support. These tasks contribute different information to greenwashing risk assessment. A model may identify climate language or vague commitments without establishing whether a reported claim is supported. Calamai et al. emphasise this distinction in a review of greenwashing text research, alongside continuing problems with dataset availability and evaluation design [5]. The following review considers the methods, findings, and relevance of each research area.

2.1 Language models and training labels

BERT learns contextual language representations through pretraining and can subsequently be fine-tuned for classification [7]. Gururangan et al. demonstrate that continued pretraining on domain and task text improves several classification tasks [8]. This approach provides a basis for adapting general language models to specialised reporting vocabulary. Sentence-BERT addresses a different need by representing sentences as numerical vectors that can be compared efficiently [9]. Cosine similarity measures the closeness of these representations, supporting retrieval and comparison of related statements. Semantic similarity identifies related meaning, while factual support requires additional evidence.

Weak supervision reduces manual labelling effort by generating labels from rules or other available signals. Snorkel combines labelling rules and models uncertainty in the resulting assignments [10]. Label provenance remains important because a faulty rule can affect many training records. Kapoor and Narayanan show how data leakage and unsuitable evaluation design can inflate machine-learning results [11]. Together, these studies motivate documenting label sources, separating evaluation groups, checking repeated text, and verifying that the intended saved model produces the reported results.

2.2 Climate topics, commitments, and disclosure quality

ClimaText provides labelled sentences for climate-topic detection [12], while ClimateBERT adds climate-specific pretraining and evaluation tasks [2]. Bingler et al. apply ClimateBERT to disclosure under the Task Force on Climate-related Financial Disclosures (TCFD) framework, identifying vague language and selective reporting [13]. A later study examines links between *cheap talk*, or low-substance climate communication, and company outcomes such as emissions growth and reputation risk [14]. These findings connect disclosure language with performance information, while retaining the distinction between a statistical association and verification of an individual claim.

ClimateBERT-NetZero uses approximately 3,500 expert-labelled examples for target classification and question answering about net-zero and reduction commitments [15]. The study separates detection of a commitment from extraction of the target details. Vinella et al. explore greenwashing risk classification using generated labels and a fine-tuned ClimateBERT model [16]. Generated labels can expand the available data, but interpretation of the results depends on the label-generation procedure. Across these studies, topic relevance, stated commitments, communication quality, and measured progress remain distinct assessment tasks.

2.3 Report-level tools and Indian disclosure studies

ChatReport combines retrieval, language-model responses, and expert feedback to support source-linked report analysis [3]. ESGReveal uses document preparation, standard questions, retrieval, and model agents to extract structured information from ESG reports [4]. Both systems demonstrate the practical value of connecting analysis outputs to report content. Source links allow a reviewer to inspect the supporting statement and surrounding context rather than relying on an isolated model response.

Within the Indian context, Shankar and Xu describe expert-labelled sustainability text and a supervised BERT classifier for sentence-level greenwashing detection [17]. The source is an SSRN working paper posted in 2025, with a manuscript dated 2024. Garg et al. examine BRSR disclosures from 1,012 businesses for 2022–23, providing reporting context and evidence on disclosure practices [6]. The institutional study addresses reporting patterns rather than a machine-learning benchmark. Table 1 summarises the reviewed studies in year order and identifies the main takeaway from each contribution.

Table 1. Selected research on language modelling and sustainability disclosure analysis. The final column summarises the takeaway from each study.

Year	Objective	Methodology	Main findings	Scope / limitation	Inference drawn
2019	General language representations: BERT [7].	Pretraining with context on both sides of a word, followed by labelled training.	One pretrained model supports several language tasks.	General pretraining does not define BRSR categories.	A pretrained encoder provides a general language base that can be adapted to a reporting task.
2019	Sentence similarity [9].	Paired sentence encoders and vector similarity.	Supports efficient search for related sentences.	Similar wording does not prove factual support.	Sentence representations support efficient comparison, while evidence support remains a separate question.
2020	Language-model domain adaptation [8].	Further pretraining on domain and task text.	Improves several reported classification tasks.	Gains depend on the training text and target task.	Specialised language can be learned before training for the final labelled task.
2020	Climate-topic detection: ClimaText [12].	Dataset with sentence-level climate labels.	Provides data for testing topic detection.	A topic label does not assess a claim's credibility.	Climate relevance provides a useful first filter, while claim assessment requires additional evidence.
2021	Climate-language modelling: ClimateBERT [2].	Further pretraining on climate text.	Improves the reported climate-language tasks.	Climate tasks do not cover every social or governance principle.	Domain adaptation improves reported climate tasks; broader reporting categories require task-specific evaluation.
2022	Climate-risk disclosure analysis [13].	ClimateBERT analysis of TCFD-related language.	Finds vague claims and selective reporting.	TCFD categories differ from BRSR principles.	Disclosure categories depend on the reporting framework used to organise the text.
2023	Net-zero and reduction-target extraction [15].	Expert labels, ClimateBERT, and question answering.	Finds targets and related details.	Detecting a target does not verify delivery.	A detected target describes a commitment; progress requires evidence about the reported outcome.
2023	Report-review support: ChatReport [3].	Text retrieval, model responses, and expert feedback.	Links report analysis to source text.	Retrieved text and answers still need evaluation.	Source-linked responses make automated report analysis easier to inspect.
2024	Risk classification with generated labels [16].	Generated labels and ClimateBERT fine-tuning.	Demonstrates an early language-model approach.	Results depend on the generated-label rules.	Generated labels can expand training data, but the label-generation method shapes the evaluation reference.
2024	Cheap talk and company outcomes [14].	Text-based indicator and outcome analysis.	Finds links with emissions growth and negative coverage.	A statistical link does not verify every claim.	Outcome data adds context to language indicators without verifying every individual statement.
2025	ESG data extraction: ESGReveal [4].	Document preparation, standard queries, retrieval, and model agents.	Supports structured extraction and disclosure analysis.	Results depend on documents and model choice.	A connected extraction workflow can organise report content into reusable records for analysis.
2025	Indian sustainability-text classification [17].	Expert-labelled text and supervised BERT.	Reports sentence-level greenwashing detection.	Working paper with different data and labels.	Supervised classification is applicable to Indian disclosures, with performance tied to the selected data and labels.

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Table 1 continued

Year	Objective	Methodology	Main findings	Scope / limitation	Inference drawn
2025	Review of Indian BRSR disclosures [6].	Institutional review of 1,012 businesses for 2022–23.	Describes BRSR reporting practices.	A disclosure study, not a detection test set.	Indian reporting practices provide context for interpreting sector and company disclosure patterns.
2025	Survey of greenwashing text research [5].	Survey of tasks, data, and methods.	Identifies separate tasks and testing gaps.	No single language task establishes greenwashing.	Topic detection, commitment extraction, and evidence assessment answer different review questions.

2.4 Research findings and motivation

The literature supports combining language analysis with source-linked evidence, documented labels, and interpretable company comparisons. GRASP applies these requirements to structured Indian BRSR reports through a shared database, a nine-principle classifier, review checks, and a four-component Transparency Score. The database preserves reporting context, the label records identify the basis of each principle assignment, and the score breakdown explains company differences. Evaluation by label source, company group, and excluded reporting concept provides complementary views of classifier behaviour. This combination defines the role of GRASP within the BRSR review workflow.

3 Proposed Solution

GRASP organises disclosure review around the structured BRSR report. Narrative text segments and reported facts are stored with the same company and reporting-period identifiers, creating a common source for classification, review checks, and scoring. This design allows an analyst to move from a company summary to the disclosure behind a result. The approach therefore addresses the central review problem identified in Section 1: connecting written claims, filed evidence, and comparable company-level outputs without losing the reporting context.

The architecture separates model training from report analysis. Report-derived weak labels support training a domain-adapted BERT classifier for the nine principles. During analysis, the trained classifier organises text segments and produces principle probabilities. Word lists and sentence similarity identify potentially vague wording, while numerical and assurance rules generate separate evidence-linked review records. The Transparency Score combines four disclosure measures and reuses the vagueness detector for Clarity, which carries a weight of 20%. Principle probabilities and numerical flags are presented alongside the score rather than included as score components. Figure 1 shows the relationships between these functions; Section 4 defines the calculations.

Implementation requires the collected reports, a database, pretrained language models, and computing resources for model training and report analysis. The trained classifier can then be reused by the service layer, which exposes company records, principle summaries, review flags, scores, and peer comparisons through an application programming interface (API). These outputs support examination of one company, selected companies in a sector, the complete financial-sector group, and all recorded sectors. Where two annual reports are available, the same scoring rules also support year-wise comparison. Evaluation in Section 5 distinguishes model agreement, language-screening performance, score comparisons, and service verification.

4 Methodology

Figure 1 presents the seven-module workflow. Report collection and text preparation support model training, while the stored report data supports principle mapping, review checks, and transparency scoring. The following subsections explain the data processing, decision rules, and calculations. The service layer brings the resulting company and sector outputs together for inspection.

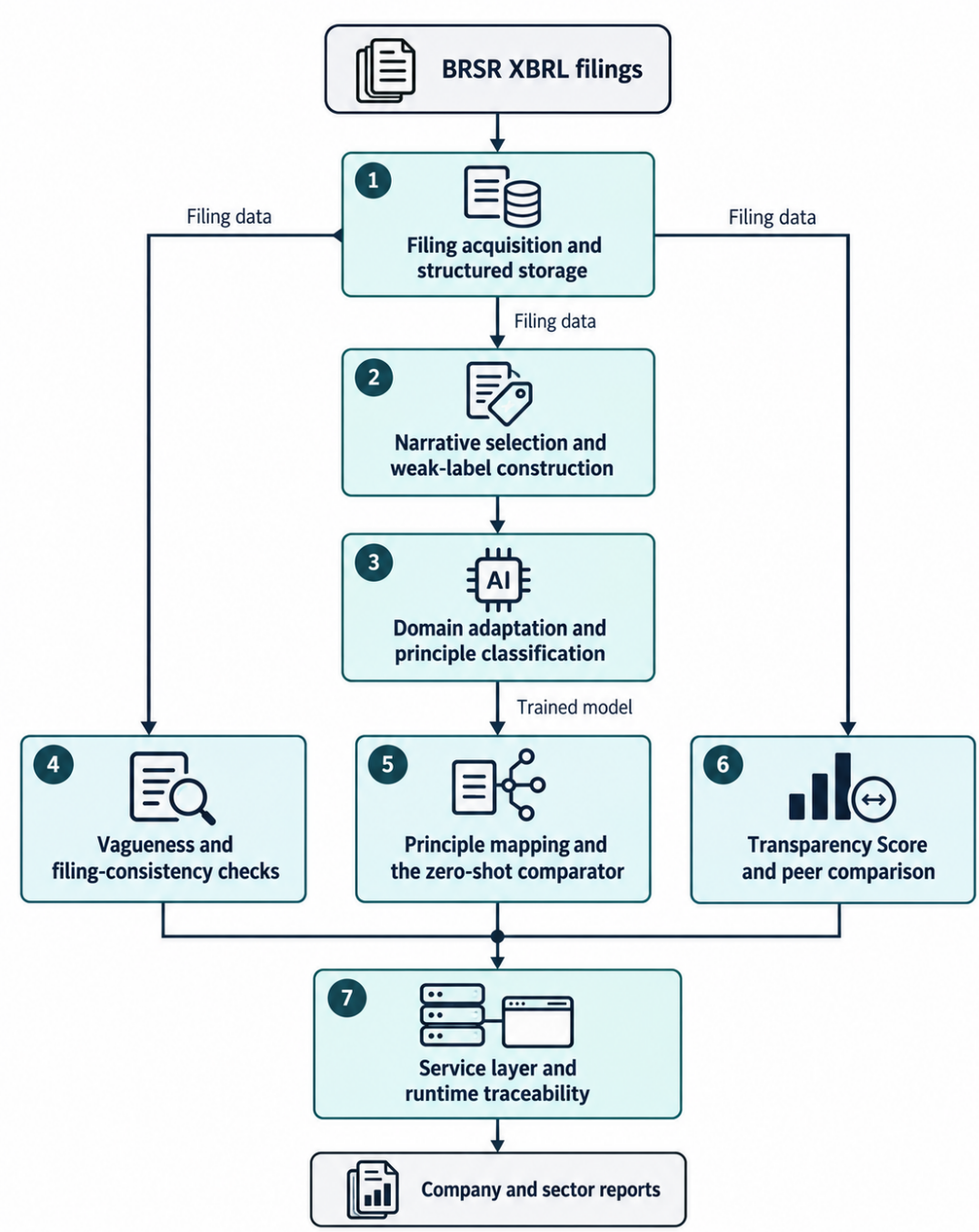


Figure 1. This figure shows the GRASP workflow from structured BRSR reports to evidence-linked company and sector assessments.

4.1 Report acquisition and structured storage

BRSR reports are collected in eXtensible Business Reporting Language (XBRL), a digital format that associates reported values with machine-readable labels and context. A download log records completed and failed files so that collection can resume after an interruption. The parser reads the structured records directly. A *concept* names a reporting item, such as total water withdrawal; the *context* identifies the company, period, and reporting categories; and the *unit* specifies the measurement scale [18]. Preserving these fields allows later checks to identify which figures are relevant to a claim.

The database links company, report, and fact records. Company records store the Corporate Identity Number (CIN), name, and industry code. Report records store the company link, reporting dates, and source location. Fact records preserve the concept, original value, unit, stated precision, period, and dimensions. Dimensions describe categories such as gender or employment type, allowing category-specific values to remain separate. Values are retained as text and converted to numbers when arithmetic is required. A company–year uniqueness constraint prevents duplicate report ingestion. The resulting database supplies the source records for text preparation, numerical checks, and score calculation.

4.2 Text segment selection and weak-label construction

Candidate text segments are extracted from concepts containing `ExplanatoryTextBlock`, which store narrative reporting responses. The builder removes URLs and retains a candidate only when at least 15 useful words remain. Bare links, short fragments, and configured non-answers are excluded. Each retained text segment receives a principle label from three sources in priority order: an explicit `PrincipleAxis` tag in Section B, a Section C concept-to-principle mapping, or a numbered `NotesPrincipleX` concept. The Section C extension contains 29 AI-assisted assignments without independent expert verification. Existing configured mappings take priority where mapping sources overlap.

Candidates are grouped by company, concept, and cleaned text. Repeated entries carrying the same label become one dataset row, meaning one text segment stored with a principle assignment. Groups carrying several labels are excluded from the single-label dataset; these groups include cross-principle responses and generic text reused under different questions. Every row retains the label source for evaluation, while only the text enters the classifier. The resulting weak labels serve as the training and evaluation reference, so classifier agreement measures consistency with the report-derived assignments.

The final dataset contains 42,409 labelled text segments. Splitting by CIN assigns 29,834 text segments from 814 companies to training, 6,374 from 175 companies to validation, and 6,201 from 175 companies to testing. No company appears in more than one supervised split. Exact text shared by different companies is assessed separately. Company scoring uses a different text-selection rule, described in Section 4.6; the classifier and scoring counts therefore refer to different retained text groups.

4.3 Domain adaptation and principle classification

The classifier is based on `bert-base-uncased`, a general-English BERT model whose tokenizer treats uppercase and lowercase forms alike [7]. Development first adapts the encoder, the part of BERT that represents input meaning, through *masked language modelling*. Selected word pieces are hidden in BRSR text and predicted from the surrounding context. This step uses unlabelled text to learn reporting vocabulary before principle classification [8]. The BERT architecture is retained throughout; domain adaptation changes the learned parameters rather than replacing the model.

Supervised fine-tuning replaces the word-prediction output with a nine-output classification layer and updates both the encoder and the new layer using the labelled training text segments. The tokenizer converts word pieces, called *tokens*, into numerical identifiers. Inputs are truncated at 512 tokens, and dynamic padding adds only the padding required within each batch. An attention mask distinguishes text positions from padding. Company identifiers, concept names, and label-source fields support grouping and evaluation but are not classifier inputs.

Training runs for five epochs, with one epoch representing a complete pass through the training group. Batches contain four text segments, and gradient accumulation over four batches gives an effective batch size of 16. The learning rate is 2×10^{-5}, with 300 warmup steps followed by cosine decay; weight decay is 0.01. Weighted cross-entropy penalises low probability for the recorded label. Class weights use inverse training frequencies, square-root smoothing, and rescaling to a mean of one, increasing the contribution of less frequent principles without applying the full inverse-frequency difference.

Validation macro-F1 selects the checkpoint, or saved parameter set, after each epoch. Five seeds—42, 1, 7, 99, and 2024—were compared on the same training and validation groups. A seed controls random choices such as data order and classification-layer initialisation. Seed 99 supplied the selected `checkpoint-7460`. LayerNorm keeps internal values on a stable scale. Recovery mapped 50 LayerNorm parameters from the older `gamma`/`beta` names to `weight`/`bias`, required all parameters to load, and verified every parameter value after saving and reloading. Recovery involved no further training. Section 5.4 presents the recorded training history and recovered-model results.

For a text segment x, the classifier produces nine raw scores, or *logits*. Softmax converts logits $z_1, \ldots, z_9$ into probabilities:

$$p_k(x) = \frac{\exp(z_k)}{\sum_{j=1}^{9} \exp(z_j)}. \tag{1}$$

Here, $p_k(x)$ is the model probability for principle k, and the nine probabilities sum to one. The highest probability gives the Top-1 prediction, while the two highest give the Top-2 candidate list. The recovered BERT checkpoint supplies the final classifier and the service predictions. MPNet supports semantic comparison in other modules; historical ClimateBERT and ensemble experiments are not part of the final classifier.

4.4 Vagueness detection and report-consistency checks

This module combines a language branch for vagueness detection with numerical rules for claim–figure, assurance-gap, and omission checks. The language branch uses both wording and evidence markers to identify text segments for review. The same detector supplies Clarity in Section 4.6. Numerical checks produce separate review records containing the relevant disclosure and reported values; these records do not enter the Transparency Score.

4.4.1 Hedging words and evidence markers

The lexical detector searches text segments for promises, conditional commitments, unclear time expressions, and unmeasured progress claims. Evidence patterns recognise percentages, currency amounts, physical quantities, workforce or incident counts, reporting years, named standards, and selected formal governance systems. A matched pattern sets the `has_kpi` marker, which records the presence of an evidence indicator rather than verifying the underlying claim. For example, *emissions fell by 12% in FY2025–26* contains a percentage and reporting year, whereas *the company aims to improve water use in the coming years* contains hedging without a reported measure.

The lexical rule flags hedging without an evidence marker. For text segments longer than 80 words, hedging density must also reach 0.005, where density is the number of matches divided by the word count. A separate rule flags bare URLs and configured website redirects. Currency matches require digits, and the pattern set includes MW/GW units and statutory terminology. The resulting hedging, evidence, and URL indicators support the final language decision; the evidence marker also supplies Specificity in the scoring module.

4.4.2 Comparing sentence meanings

The `all-mpnet-base-v2` encoder represents sentences as vectors, and cosine similarity compares each sentence with fixed vague and specific reference statements [9, 19]. For sentence j, let v_j be the highest similarity to a vague reference and q_j the highest similarity to a specific reference. The detector selects the sentence with the largest $q_j - v_j$, representing the strongest specific wording within the text segment, and applies the decision thresholds to that sentence.

Without an evidence marker, the semantic rule flags a text segment when $v_j \geq 0.42$ and $v_j > q_j$. With a marker, flagging requires $v_j \geq 0.58$ and $v_j - q_j \geq 0.15$. A URL-redirect match takes priority. The final language flag is raised when either the lexical or semantic rule applies. A saved freeze manifest records the detector version and thresholds, which remain unchanged during the reported evaluations. The output identifies potentially vague text for inspection rather than confirming misleading conduct.

4.4.3 Claim–figure, assurance, and omission checks

The claim–figure rule covers emissions, water, waste, and non-renewable energy. A sentence must contain both improvement wording and a phrase associated with the measure. An increase in the corresponding filed value creates a review record. Matching the improvement and measure within the same sentence reduces unrelated pairings, such as linking a Scope 2 claim with a Scope 1 figure. The record preserves the sentence, concept, previous and current values, and assurance information. Analysts can then check measure definitions, totals versus intensity values, units, periods, and business boundaries against the original report.

The assurance-gap rule connects a domain-specific fact with assurance information for the same domain. The omission rule identifies a concept reported in the previous period without a matching current-period value. These rules retain the implementation's matching conditions: assurance selection does not fully restrict period, unit, and dimension, and zero-valued entries receive different treatment across rules. The assurance rule treats a text zero as missing, while the omission rule skips some previous zero entries. In contrast, Completeness accepts structured zero values. These differences define how review records are generated and why the linked source values remain available. The evaluation of detector accuracy covers the language branch, as specified in Section 5.

4.5 Principle mapping and the zero-shot comparator

Principle mapping reuses the recovered classifier to summarise support for the recorded principle assignments in a report. For text segments filed under principle k, the service averages the corresponding probabilities from Equation (1) and retains the lowest-scoring text segment for inspection:

$$\bar{p}_k = \frac{1}{N_k} \sum_{x \in X_k} p_k(x), \tag{2}$$

Here, X_k is the set of text segments assigned to principle k, and $N_k = |X_k|$ is the number of text segments in that set. The company summary averages the available principle means, excluding

principles without a text segment. These values describe model support for the recorded assignment rather than a legal compliance percentage.

A zero-shot MPNet method provides the evaluation comparator without fitting to the principle-labelled training set. Each principle has an expanded definition and fixed example statements, called *anchors*. The ranking score is

$$z_k(x) = 0.50\, d_k(x) + 0.50 \left[0.70 \max_a s_{ka}(x) + 0.30\ \text{mean}_a\, s_{ka}(x)\right]. \tag{3}$$

In this equation, $d_k(x)$ measures similarity between text segment x and the definition of principle k, while $s_{ka}(x)$ measures similarity to anchor a. The definition and anchor terms each receive half the weight. Within the anchor term, the best match receives 70% and the mean similarity receives 30%. Section 5.7 compares both methods on the same test text segments and weak-label reference.

4.6 Transparency Score and peer comparison

The Transparency Score summarises report coverage and disclosure characteristics through four components on a 0–100 scale:

$$TS_f = 0.40\, C_f + 0.30\, Sp_f + 0.20\, Cl_f + 0.10\, G_f. \tag{4}$$

For report f, C_f is Completeness, Sp_f is Specificity, Cl_f is Clarity, and G_f is SDG breadth. The implementation fixes the weights at 0.40/0.30/0.20/0.10 and rounds each component and the combined score to two decimal places. Report facts supply coverage, retained text segments supply evidence and vagueness counts, and a fixed principle map supplies SDG breadth. The following definitions specify the numerator and denominator of each component.

4.6.1 Completeness: coverage of reporting items

Let U be the database-wide set of 1,057 distinct concepts, including structured and narrative items. For report f, let A_f contain concepts with at least one accepted response. Completeness is

$$C_f = 100\, \frac{|A_f|}{|U|}. \tag{5}$$

A concept counts once regardless of the number of dimensional values. The same denominator of 1,057 applies to every report, including concepts used by other sectors; the measure therefore represents coverage of a fixed database checklist. Blank values and placeholders such as *NA* and *not applicable* are rejected, while structured zero is accepted. Narrative text blocks reject placeholders and lone zero values. Text longer than 30 characters must contain at least ten words; shorter values are accepted when no placeholder matches. Address fields follow a separate acceptance rule.

4.6.2 Specificity: presence of evidence markers

Scoring retains text-block facts containing at least 30 characters and removes duplicate concept–text pairs after trimming extra spaces. Identical wording under different concepts may therefore count separately. Let N_f be the retained text-segment count and K_f the count containing at least one evidence marker from Section 4.4. Specificity is

$$Sp_f = 100\, \frac{K_f}{N_f}. \tag{6}$$

The numerator counts text segments with a recognised marker, not individual numbers or standards. A text segment containing several measurements contributes once. This selection rule differs from the classifier's minimum of 15 useful words, so scoring and classification use different text groups.

4.6.3 Clarity: absence of a vagueness flag

Clarity reuses the detector from Section 4.4, with the same word lists, semantic thresholds, and URL rules. Let V_f be the number of flagged scoring text segments. Using the same denominator N_f as Specificity,

$$Cl_f = 100\left(1 - \frac{V_f}{N_f}\right). \tag{7}$$

A high Clarity value indicates that few retained text segments were flagged as vague. Concrete evidence is measured separately by Specificity, so a report can have high Clarity and low Specificity. The implementation returns zero for both components when no scoring text segments are available. Clarity contributes 20% of the combined score; claim–figure, assurance-gap, and omission records remain separate review outputs.

4.6.4 SDG breadth: coverage under a fixed principle map

SDG breadth measures how many of the 17 United Nations Sustainable Development Goals are reached through a fixed mapping from the nine BRSR principles [20]. A principle qualifies when an explicitly tagged narrative has at least 20 characters and contains an evidence marker. Let D_f contain the qualifying principles, and let $M(k)$ denote the goal numbers assigned to principle k. The fixed mapping is

$$\begin{aligned}
M(1) &= \{16\}, & M(2) = M(9) &= \{9, 12\},\\
M(3) &= \{3, 5, 8\}, & M(4) &= \{10, 17\},\\
M(5) &= \{8, 10, 16\}, & M(6) &= \{6, 7, 12, 13, 15\},\\
M(7) &= \{16, 17\}, & M(8) &= \{1, 8, 10, 11\}.
\end{aligned}$$

The percentage of distinct mapped goals is

$$G_f = 100\,\frac{\left|\bigcup_{k \in D_f} M(k)\right|}{17}. \tag{8}$$

The union counts each goal once, including a goal reached through several principles. The fixed map covers 14 goals, giving a maximum of 82.35% against the denominator of 17. This component measures mapped breadth rather than achievement of individual SDG targets. Saved exports use the name *SDG Alignment*; the paper uses *SDG breadth* to describe the calculation. If a duplicate concept–text pair appears under several principles, database record order determines the retained principle.

4.6.5 Sector grouping and percentiles

Companies are assigned to recorded sectors using industry codes and a local mapping. The analysis retains Financial Services (excl. Insurance) as a separate group from insurance and auxiliary financial services. For company c in sector S, the percentile is

$$P_c = 100\,\frac{\#\{j \in S : TS_j < TS_c\}}{|S|}. \tag{9}$$

The numerator counts sector companies with a strictly lower Transparency Score, and $|S|$ is the total sector size. Equal scores receive equal percentiles, and the highest percentile need not reach 100. The main sector comparison includes groups with at least five companies and reports the mean, median, range, and standard deviation of company scores. The service returns the combined score, component percentages, and sector percentile so that a company comparison can be traced to the underlying calculation.

4.7 Service layer and runtime traceability

An API exposes company search, report analysis, score breakdowns, peer comparisons, and principle mapping using the stored records and analysis outputs. The principle-mapping response identifies the method as `recovered_seed99_supervised_classifier` and returns the recovered checkpoint path. An unavailable classifier produces an error rather than a silently substituted method. Model-loading status, principle probabilities, review records, and peer comparisons occupy separate response fields. Section 5 reports the request used to verify this connection; service verification is distinct from model-performance evaluation.

5 Evaluation and Model Results

Evaluation covers principle classification, text-vagueness screening, comparison with historical human ratings, and verification of the service connection. Each result identifies the text group or company cohort and the reference used for comparison. Classifier and mapping results use the recovered seed-99 checkpoint. Company-score applications are presented separately in Section 6.

5.1 Evaluation measures and uncertainty

Top-1 agreement is the proportion of text segments whose highest-ranked principle matches the recorded weak label. Top-2 agreement counts a match when the label appears among the two highest-ranked principles. Precision describes how often a predicted principle matches the reference, while recall describes how much of a reference category is recovered. For principle k, F1 combines precision P_k and recall R_k through the harmonic mean:

$$F1_k = \frac{2P_kR_k}{P_k + R_k}, \qquad \text{MacroF1} = \frac{1}{9}\sum_{k=1}^{9} F1_k. \tag{10}$$

Macro-F1 averages the nine principle-level F1 values with equal weight, preventing the largest class from dominating the measure. Support is the number of reference text segments for a principle. All classification measures assess agreement with report-derived labels rather than independent semantic judgements.

Classification and mapping intervals use 2,000 company-cluster bootstrap samples with seed 4206. Companies are drawn with replacement, retaining all text segments from each selected CIN, and paired comparisons use the same draw for both methods. The 20-company score comparison uses 10,000 bootstrap samples of company-score pairs with seed 42. The resulting 95% confidence intervals describe sampling variation within the available data. Detector confidence intervals were not calculated. The supervised split has no shared companies, but the earlier masked-language-model builder reads across the company database, and the exact adaptation-corpus membership of the selected encoder is unverified. Company-disjoint claims therefore concern supervised fine-tuning. The reported recovery results re-evaluate the established test group rather than a newly collected cohort.

5.2 Comparative evaluation

Table 2 brings together three complementary comparisons: label sources, excluded reporting concepts, and supervised versus zero-shot ranking. Each comparison addresses a different question and retains the relevant data and training conditions. Domain adaptation is part of the final training sequence, but these comparisons do not isolate a numerical gain from adaptation or data cleaning.

Table 2. Overview of classifier comparisons. Each row states the evaluation group and training conditions; the rows do not form a single-component ablation.

Comparison	Evaluation population	Top-1 agreement	Interpretation
Label source	B: 352; C: 5,364; notes: 485 text segments	B: 63.35%; C: 95.15%; notes: 90.72%	Results for parts of the same 6,201-text-segment test group. The sources use different text and labelling rules.
Full-concept and excluded-concept training	Same 1,190 text segments from 174 companies	Full concept: 97.98%; excluded concept: 87.31%	The two training sets differ in concept coverage, number of training examples, and seed.
Zero-shot MPNet versus supervised BERT	Same 6,201 text segments from 175 companies	MPNet: 67.63%; BERT: 93.00%	The two methods have different access to labels. The BERT value reuses the classifier result.

5.3 Dataset composition and label-source results

The dataset contains 42,409 labelled text segments: 2,479 from explicit principle-axis facts, 36,513 from Section C mappings, and 3,417 from principle notes. Filtering excludes 36,491 noisy facts and removes 2,640 groups carrying several labels. These counts describe different units and cannot be added as discarded final text segments. Within Section B, 41,898 narrative candidates reduce to 25,214 after removing 16,684 unsuitable entries. Removing 18 same-label duplicates and 22,717 entries from multi-label groups leaves 2,479 dataset rows. Table 3 reports the final principle distribution and test counts.

Table 3. Distribution of labelled text segments across the nine principles and the company-disjoint test group.

Principle	All text segments	Test text segments
P1: Ethics and governance	5,015	773
P2: Sustainable goods and services	5,566	808
P3: Employee and worker wellbeing	11,835	1,715
P4: Stakeholder interests	3,128	469
P5: Human rights	3,095	449
P6: Environmental responsibility	5,749	823
P7: Public policy advocacy	543	69
P8: Inclusive growth	2,475	368
P9: Consumer responsibility	5,003	727
Total	**42,409**	**6,201**

Source-wise agreement varies substantially, as shown in Table 4 and Figure 2. Top-1 agreement is 63.35% for Section B, 95.15% for Section C, and 90.72% for principle notes. Section C contributes 86.1% of the full dataset and 5,364 of the 6,201 test text segments, so the overall result is strongly influenced by this source. Section B results describe the subset retained after single-label filtering. Reporting source-wise performance alongside the overall measure makes these differences visible.

Table 4. Agreement by label source within the same test group. Company counts may overlap across sources; intervals resample companies.

Source	Text segments	CINs	Top-1	Top-2	Top-1 95% CI
Section B: explicit axis	352	71	63.35%	80.40%	56.55–69.67%
Section C: concept map	5,364	174	95.15%	99.01%	94.41–95.92%
Principle notes	485	103	90.72%	95.46%	86.55–94.09%

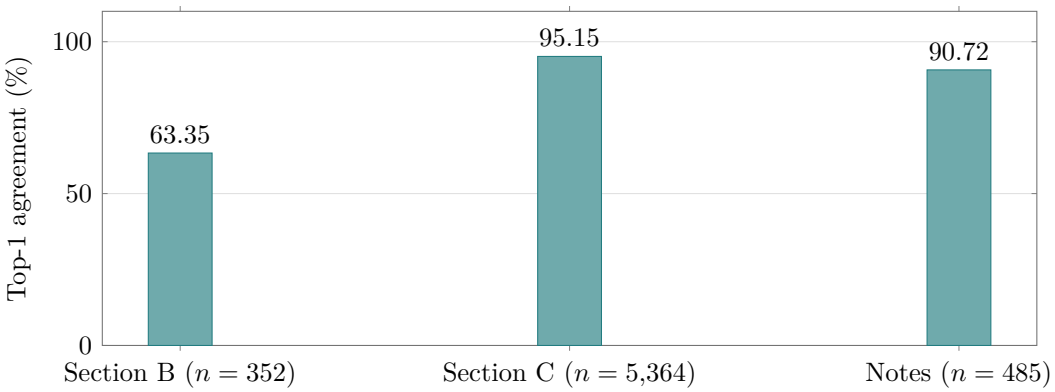
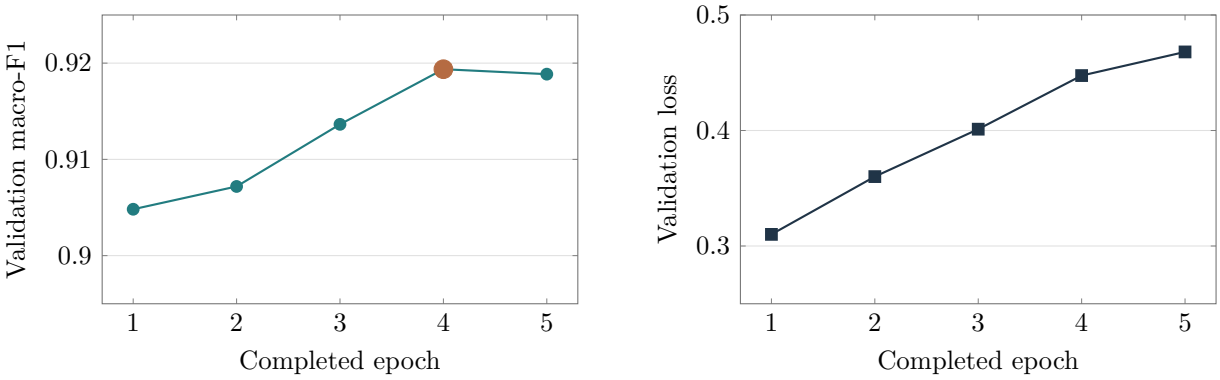


Figure 2. Top-1 agreement for explicitly tagged text segments, concept-mapped text segments, and principle notes. Company-cluster confidence intervals appear in Table 4.

5.4 Classifier performance and training history

The recovered classifier matches 5,767 of 6,201 test labels, giving 93.00% Top-1 agreement (95% CI: 92.04–93.89%). The reference label appears among the two highest predictions for 6,057 text segments, giving 97.68% Top-2 agreement (95% CI: 97.10–98.21%). Macro-F1 is 0.9158 (95% CI: 0.9036–0.9277). Although the supervised splits share no CINs, 50 test text segments exactly match training wording from other companies, accounting for 0.81% of the test group.

Figure 3 shows the five recorded epochs for seed 99. Validation macro-F1 increases from 0.9048 at epoch 1 to 0.9194 at epoch 4, followed by 0.9189 at epoch 5. The selection criterion therefore retains epoch 4. Validation loss rises over the same period because loss also reflects prediction confidence, whereas macro-F1 measures the balance of category decisions. Table 5 and Figure 4 report performance across all nine principles. P7 has 69 test text segments, compared with 1,715 for P3, making class support important when interpreting the individual scores.



(a) Validation macro-F1 peaks at epoch 4.

(b) Validation loss increases across epochs.

Figure 3. Validation history for seed 99 across five epochs. The marked point identifies checkpoint 7,460, selected by macro-F1 rather than minimum validation loss.

Table 5. Per-principle results against weak labels on 6,201 test text segments. Support gives the reference count; precision, recall, and F1 describe category performance.

Principle	Support	Precision	Recall	F1
P1: Ethics and governance	773	0.9592	0.9431	0.9511
P2: Sustainable goods and services	808	0.9154	0.8837	0.8992
P3: Employee and worker wellbeing	1,715	0.9533	0.9644	0.9588
P4: Stakeholder interests	469	0.9611	0.9488	0.9549
P5: Human rights	449	0.9009	0.8508	0.8751
P6: Environmental responsibility	823	0.8943	0.9356	0.9145
P7: Public policy advocacy	69	0.8551	0.8551	0.8551
P8: Inclusive growth	368	0.8921	0.9212	0.9064
P9: Consumer responsibility	727	0.9259	0.9285	0.9272
Macro average	6,201	0.9175	0.9146	0.9158

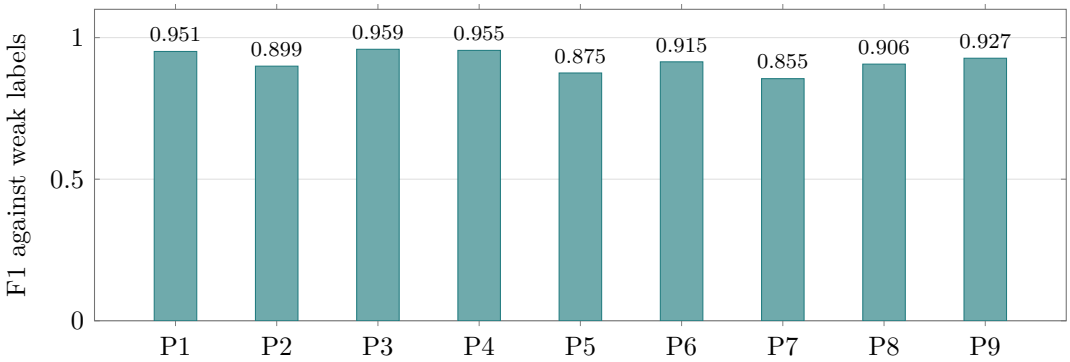


Figure 4. F1 across the nine principles. Higher values indicate a better balance of precision and recall against weak labels. Table 5 provides the corresponding text-segment counts.

5.5 Generalisation to excluded reporting concepts

Company separation tests new supervised company groups, while concept exclusion tests reporting questions omitted from labelled training. Excluding 17 Section C concepts produces an evaluation

group of 1,190 text segments. On this same group, the full-concept model achieves 97.98% Top-1 agreement and the excluded-concept model achieves 87.31%. The paired difference is −10.67 percentage points (95% CI: −12.36 to −8.96). Both checkpoints passed strict parameter-name recovery. The full-concept model uses seed 99, whereas the excluded-concept model uses seed 42 and fewer training text segments; the observed difference therefore reflects these conditions together.

Across three excluded-concept runs, mean Top-1 agreement is 81.64%, with a standard deviation of 4.92 percentage points. Mean Top-2 agreement is 95.34% (standard deviation 1.31 percentage points), and mean macro-F1 is 0.6752 (standard deviation 0.0170). Figure 5 shows the individual results and evaluation sizes. These standard deviations describe variation across the three runs, not confidence intervals. All runs reuse the original test-company group, and concept exclusion applies to labelled training; the adaptation-corpus condition described above still applies.

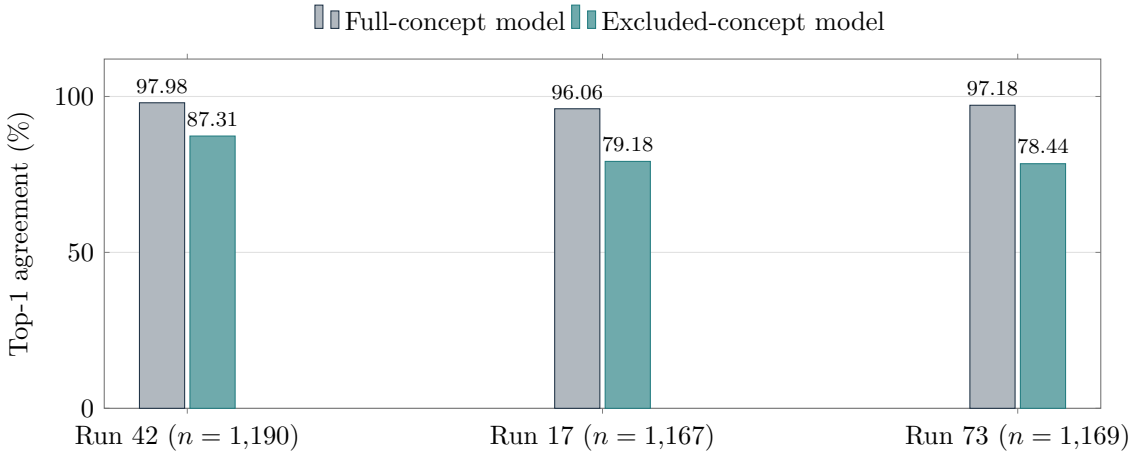


Figure 5. Concept-exclusion results across three evaluation groups containing 174, 173, and 174 companies. The excluded-concept models achieve mean Top-1 agreement of 81.64%, with standard deviation 4.92 percentage points. Training conditions differ as described in Section 5.5.

5.6 Fixed text-vagueness detector

Table 6 reports two separate label groups under fixed detector settings. The development group contains 80 text segments, with 37 labelled vague. The detector raises 41 flags, of which 29 match the reference. Precision is therefore $29/41 = 0.7073$, and recall is $29/37 = 0.7838$. Sixty decisions agree with the reference, giving accuracy of 0.7500 and F1 of 0.7436. Blinding of the earlier development labels is not documented, so this result is reported as a development comparison.

The second group contains 70 AI-annotated text segments. The detector raises 35 flags, with 21 matching the 26 AI-vague labels, giving F1 of 0.6885. Figure 6 compares precision, recall, and F1 across the two groups. The AI comparison is exploratory rather than independent human validation, and the groups are not pooled because the text and reference sources differ.

Table 6. Language-detector performance under fixed settings. Development labels and AI annotations are separate evaluation references.

Reference labels	N	Accuracy	Precision	Recall	F1
Historical development labels	80	0.7500	0.7073	0.7838	0.7436
AI annotations	70	0.7286	0.6000	0.8077	0.6885

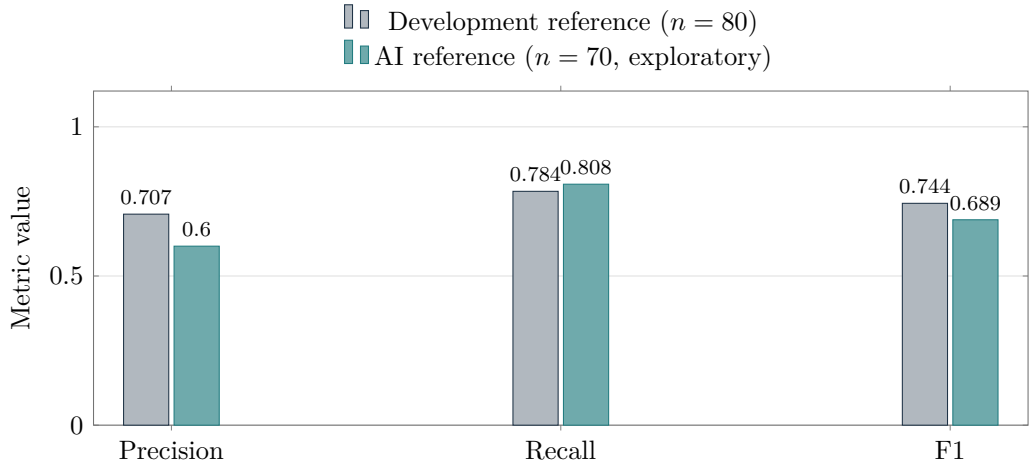


Figure 6. Precision, recall, and F1 for the fixed language detector against 80 development labels and 70 AI annotations. The two references describe separate comparisons and are not pooled.

5.7 Supervised transfer versus zero-shot mapping

Both ranking methods are evaluated on the same 6,201 text segments and report-derived principle labels. MPNet's definition-plus-anchor method achieves 67.63% Top-1 and 83.34% Top-2 agreement, compared with 93.00% and 97.68% for the recovered supervised classifier. Table 7 gives the corresponding confidence intervals and macro-F1 values. The paired Top-1 difference is +25.37 percentage points (95% CI: +24.02 to +26.71). This comparison includes differences in representation and access to labelled training, so the gap is not an isolated measure of encoder quality. The supervised row reuses the classifier evaluation rather than providing a second independent test.

Table 7. Supervised and zero-shot principle ranking on the same company-disjoint test group. Confidence intervals use company-cluster resampling.

Method	Top-1	Top-2	Macro-F1
MPNet zero-shot hybrid	67.63%	83.34%	0.6405
95% interval	66.45–68.93%	82.31–84.38%	0.6248–0.6567
Recovered supervised transfer	93.00%	97.68%	0.9158
95% interval	92.04–93.89%	97.10–98.21%	0.9036–0.9277

5.8 Transparency Score and historical human ratings

Across 20 historical company ratings, Pearson correlation between automated and human scores is $r = 0.6314$ (95% CI: 0.2683–0.8519; $p = 0.0028$). Spearman rank correlation is $\rho = 0.6285$ (95% CI: 0.2321–0.8540). Mean absolute error is 12.16 points, and root mean squared error is 15.19 points. The squared correlation is $r^2 = 0.3986$, describing association without a separately fitted predictive regression. These measures report both how the scores vary together and the size of the score differences.

All 20 ratings come from one documented source, and rater blinding is not recorded. The original Zensar packet contains a URL-related extraction issue, while Geojit's business model differs from manufacturing-oriented reporting items. Both companies remain in the reported cohort. The comparison retains the original ratings and fixed score formula; synthetic ratings and AI assessments are separate records.

5.9 API model-loading verification

A principle-mapping request for 360 ONE WAM LIMITED on 19 October 2026 returned the recovered seed-99 method identifier and checkpoint path. The response included a mean principle probability of 89.17% for that report. This request verifies the service connection to the intended model for one company. The returned probability summarises the report and is not an additional accuracy measurement.

6 Company, Sector, and Year-wise Results

The application results present four reporting levels: one company, selected companies within a sector, the complete financial-sector group, and all recorded sectors. A separate paired-year analysis examines change over time. The database contains 1,164 companies and 1,172 reports, including eight companies with two reports. Company and sector summaries use one score per CIN from the latest available report selected by the export procedure. The following examples connect these summaries with the component counts and source records.

6.1 One company: evidence and score calculation

Triveni Turbine Limited provides a worked example with complete score components and a sector comparison. The report ends on 31 March 2026 and contains 1,862 raw facts. Table 8 presents previous- and current-year values for matched concepts and units. Scope 1 emissions increase, whereas Scope 2 emissions and water withdrawal decrease. The unit tCO2e means tonnes of carbon dioxide equivalent. These different directions illustrate why a broad claim about environmental improvement needs a measure-specific check, including the reporting period and business boundary.

Table 8. Previous- and current-year values from Triveni Turbine report record 197. The original concept, unit, and period fields remain linked to each value.

Filed measure	FY2024–25	FY2025–26	Filed unit
Scope 1 emissions	4,142.84	5,177.03	tCO2e
Scope 2 emissions	3,353.61	3,169.15	tCO2e
Total water withdrawal	88,662.28	85,611.54	Kilolitres

The score can be reconstructed from the retained records. Completeness is $100(613/1057) = 57.99\%$, based on 613 accepted concepts. Among 52 retained concept–text pairs, 33 contain an evidence marker, giving Specificity of 63.46%. One text segment is flagged as vague, giving Clarity of 98.08%. Twelve mapped goals produce SDG breadth of 70.59%. Table 9 applies the fixed weights, yielding $23.196 + 19.038 + 19.616 + 7.059 = 68.909$, rounded to 68.91. Figure 7 shows the weighted component contributions. Within Machinery and Equipment, the company has a sector percentile of 94.4, compared with a mean score of 49.46 across 36 companies.

Table 9. Weighted components of the Triveni Turbine score. Each contribution equals the component percentage multiplied by the fixed weight.

Component	Percentage	Weight	Contribution
Completeness	57.99	0.40	23.196
Specificity	63.46	0.30	19.038
Clarity	98.08	0.20	19.616
SDG breadth	70.59	0.10	7.059
Transparency Score			**68.91**

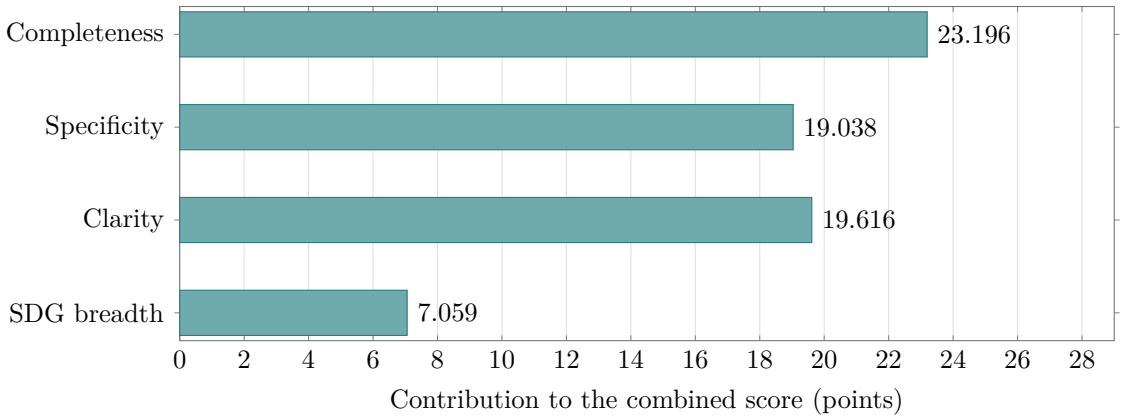


Figure 7. Weighted contributions to the Triveni Turbine Transparency Score. Each bar equals the component percentage multiplied by the fixed weight; the contributions total 68.91 after rounding.

6.2 Selected financial companies: component differences

The saved Financial Services (excl. Insurance) group contains 94 companies. Table 10 presents the four highest and three lowest scores, showing component differences across the score range rather than a random company group. REC Limited scores 67.56, compared with 29.83 for Vardhman Holdings. Specificity is 57.69% and 2.78%, respectively, while Completeness is 56.01% and 35.00%. The shared formula makes the source of the score difference visible. Business activity, applicable reporting items, and report year remain relevant when interpreting the comparison; Supplementary Table S1 retains the complete company records.

Table 10. Selected high- and low-scoring financial companies. FY end denotes 31 March of the stated year, reconstructed from the latest report per CIN. Components are percentages; TS is the combined score.

Company	FY end	*C*	*Sp*	*Cl*	*G*	TS
REC	2026	56.01	57.69	98.08	82.35	67.56
Mahindra & Mahindra Financial Services	2026	67.83	49.15	98.31	29.41	64.48
YES BANK	2026	60.45	46.67	92.00	70.59	63.64
KOTAK MAHINDRA BANK	2026	61.21	59.42	95.65	17.65	63.21
Fedbank Financial Services	2025	31.98	12.12	75.76	0.00	31.58
Nalwa Sons Investments	2025	31.13	10.26	74.36	0.00	30.40
Vardhman Holdings	2025	35.00	2.78	75.00	0.00	29.83

6.3 The complete financial-sector group

Across all 94 financial companies, the mean score is 48.17, the median is 49.79, and the standard deviation is 9.88. The median identifies the middle score, while standard deviation describes the spread around the mean. Scores range from 29.83 to 67.56, a difference of 37.73 points. Mean component values are 49.77% for Completeness, 28.35% for Specificity, 90.67% for Clarity, and 16.27% for SDG breadth. The separation between Clarity and Specificity shows that avoiding vague wording does not necessarily provide concrete evidence.

Table 11 groups companies into four score bands. Eleven companies score at least 60, while 22 score below 40. Mean Specificity rises from 14.30% in the lowest band to 47.24% in the highest. The sector mean is 0.56 points below the overall company mean of 48.73, although the company range shows substantial variation within the sector. The latest report ends in March 2026 for 77 companies and March 2025 for 17 companies. These dates were reconstructed by CIN because the original score export omitted report identifiers, so this comparison uses the latest available reports rather than one common reporting year.

Table 11. Financial-sector score distribution across all 94 companies. Component columns report band means. Complete company values and reconstructed dates appear in Supplementary Table S1.

Score band	**N**	**Mean TS**	**Range**	*C*	*Sp*	*Cl*	*G*
60.00–69.99	11	62.39	60.01–67.56	61.17	47.24	95.25	47.06
50.00–59.99	35	54.77	50.23–59.84	60.69	32.94	94.07	17.98
40.00–49.99	26	45.22	40.57–49.96	45.85	26.05	89.79	11.08
Below 40.00	22	34.05	29.83–39.80	31.33	14.30	84.02	4.28
All financial companies	94	48.17	29.83–67.56	49.77	28.35	90.67	16.27

6.4 All sectors: coverage and score distribution

The complete collection contains 1,164 company scores across 72 recorded sectors. The mean is 48.73, the median is 47.25, and scores range from 26.88 to 75.58. Forty-seven sectors contain at least five companies and account for 1,109 scores; the remaining 25 sectors contain 55 companies. Table 12 lists the 47 eligible sectors and pools the smaller groups. Supplementary Table S2 provides all 72 sector rows, including medians and group sizes.

Among eligible groups, Warehousing and Transport Support has the highest mean, 54.55 across six companies. Wearing Apparel has the lowest mean, 42.33 across six companies, followed by Education at 42.43 across six companies. The highest and lowest means differ by 12.22 points. Larger groups provide additional context: Chemicals Manufacturing contains 101 companies with a mean of 49.56, Financial Services contains 94 with a mean of 48.17, and Pharmaceuticals contains 65 with a mean of 51.18. Pharmaceuticals scores range from 32.38 to 73.72, compared with 32.18 to 75.34 in Chemicals Manufacturing. These figures describe the saved scores and show variation both within and between recorded sectors.

Table 12. Transparency Scores across all 1,164 companies. The 47 sectors with at least five companies are listed separately, and 25 smaller sectors are pooled. SD is the standard deviation. Full sector records appear in Supplementary Table S2.

Sector or reporting group	N	Mean TS	Range	SD
Chemicals Manufacturing	101	49.56	32.18–75.34	9.32
Financial Services (excl. Insurance)	94	48.17	29.83–67.56	9.88
Pharmaceuticals	65	51.18	32.38–73.72	10.06
Basic Metals	60	50.80	34.61–75.58	10.87
Electrical Equipment	51	48.80	26.88–67.03	9.56
Computer Programming and Information Technology Services	49	50.24	32.89–73.98	12.25
Non-Metallic Mineral Products	40	50.36	35.66–69.20	8.69
Food Products Manufacturing	38	47.57	30.04–69.86	9.60
Motor Vehicles	37	51.76	33.29–70.12	9.98
Machinery and Equipment	36	49.46	35.61–69.67	10.06
Rubber and Plastics	36	49.22	35.44–73.87	9.90
Auxiliary Financial Services	31	47.26	27.67–61.39	9.08
Electricity, Gas, Steam and AC Supply	31	53.67	31.95–74.23	11.49
Fabricated Metal Products	27	50.29	35.03–65.72	8.66
Textiles	26	47.44	31.77–67.65	10.16
Wholesale Trade	24	45.46	28.05–64.86	12.39
Civil Engineering	23	48.31	31.99–66.15	10.84
Retail Trade	21	43.06	34.22–56.45	6.12
Construction	20	48.73	36.58–63.25	8.32
Insurance and Pension Funding	20	51.52	29.92–64.17	10.12
Real Estate Activities	19	47.90	35.17–63.42	8.78
Human Health Activities	18	46.37	33.92–62.86	8.12
Information Service Activities	18	47.39	33.97–64.19	9.71
Computer, Electronic and Optical Products	17	47.46	32.93–65.45	7.59
Other Transport Equipment	17	50.83	37.67–66.89	9.39
Other Manufacturing	16	46.11	30.59–63.81	10.19
Telecommunications	15	50.63	36.57–69.19	11.43
Accommodation	14	48.67	31.28–67.57	10.06
Mining and Quarrying	13	47.40	29.01–73.16	13.27
Beverages Manufacturing	10	47.30	34.70–67.71	13.18
Other (Division 99)	10	48.39	34.80–73.44	12.12
Wholesale/Retail Trade - Motor Vehicles	10	49.25	37.40–67.92	11.47
Architecture and Engineering	9	48.50	37.49–64.86	10.06
Coke and Refined Petroleum	9	52.32	38.20–69.63	11.13
Paper Products	9	43.27	33.42–48.93	4.88
Broadcasting	8	43.65	31.66–52.74	7.64
Land Transport	7	47.42	34.72–63.63	10.11
Water Supply and Waste Management	7	43.33	36.73–56.76	8.29
Wood Products	7	46.18	36.16–64.51	9.72
Education	6	42.43	33.28–55.70	7.74
Furniture	6	51.28	36.17–65.29	9.25
Other Professional/Technical Activities	6	48.43	38.73–57.90	8.70
Scientific R&D	6	44.90	38.95–52.11	5.71
Warehousing and Transport Support	6	54.55	37.82–67.03	10.34
Wearing Apparel	6	42.33	37.17–48.78	4.50
Leather Products	5	50.00	42.14–58.86	7.80
Water Transport	5	48.37	38.70–61.22	9.75
Other 25 small sectors (combined)	55	44.08	27.56–70.97	8.95
All 72 sectors	**1,164**	**48.73**	**26.88–75.58**	**9.98**

Within the recorded Pharmaceuticals group, Dabur India scores 73.72 and Kopran scores 32.38, a gap of 41.34 points. Completeness is 73.13% for Dabur India and 32.92% for Kopran; Specificity is 57.38% and 23.68%, respectively. Clarity exceeds 85% for 58 of the 65 companies, so differences in coverage and evidence markers remain important even when vague wording is uncommon. Supplementary Table S4 preserves the four highest and three lowest examples with every component.

6.5 Year-wise changes in the available paired reports

Eight companies across sectors have reports for both FY2024–25 and FY2025–26. Table 13 compares the paired scores and identifies the component with the largest absolute weighted contribution to each change. Six scores rise and two fall. The group mean increases from 45.17 to 50.40, or 5.23 points. This available-case comparison describes the eight recorded pairs rather than a market-wide trend. Supplementary Table S3 retains both report identifiers, year labels, and all component values.

Table 13. Paired-year scores for eight companies across sectors. The final column identifies the largest absolute weighted contribution to each score change.

Company	FY2024–25	FY2025–26	Change	Largest contribution
Praj Industries	38.55	50.30	+11.75	*C*: +9.91
ESAF Small Finance Bank	42.59	54.10	+11.51	*C*: +8.10
Sudarshan Colorants	49.74	58.78	+9.04	*Sp*: +7.00
Gopal Snacks	41.30	47.29	+5.99	*Sp*: +8.53
G R Infraprojects	35.94	38.87	+2.93	*Sp*: +3.07
Sudarshan Chemical	62.95	65.32	+2.37	*G*: +2.35
VIP Industries	45.91	45.56	-0.35	*Cl*: -2.36
SML Mahindra	44.38	42.98	-1.40	*Cl*: -1.17
Group mean	45.17	50.40	+5.23	—

ESAF Small Finance Bank illustrates the component calculation over time. Completeness rises from 34.34% to 54.59%, Specificity from 30.23% to 41.30%, and Clarity from 93.02% to 93.48%, while SDG breadth remains 11.76%. The combined score increases from 42.59 to 54.10. Of the 11.51-point gain, Completeness contributes about 8.10 points, Specificity 3.32, and Clarity 0.09. This decomposition explains the recorded score change using the same fixed weights applied in the company and sector comparisons.

7 Conclusion

GRASP provides an evidence-linked workflow for assessing disclosure characteristics and greenwashing risk in structured Indian BRSR reports. The central contribution is the connection between report-derived principle labels, language and numerical review checks, and company scores with visible component calculations. Shared report identifiers allow an analyst to inspect the text or filed value behind a result, while company and sector views support review at different levels. The vagueness detector also supplies the Clarity component of the Transparency Score, maintaining a direct link between text screening and company-level scoring.

The recovered classifier achieves 93.00% Top-1 agreement and macro-F1 of 0.9158 against weak labels on 6,201 text segments from 175 companies excluded from supervised training. Source-wise and excluded-concept results provide further context for the overall agreement. The fixed language detector achieves F1 of 0.7436 on 80 development text segments, and the Transparency Score correlates with 20 historical human ratings at $r = 0.6314$. Application outputs cover 1,164 companies in 72 recorded sectors. The four reporting levels show how component differences explain company and sector scores, while the eight paired-year cases record a mean increase of 5.23 points. Each result remains tied to the evaluation reference and company group described in the paper.

This work can be extended using multiple levels. At the data level, collection can include annexures, charts, and additional annual reports. At the modelling level, larger labelled datasets, blind human-rating studies, and adaptation restricted to declared training companies can support further evaluation on newly collected companies. Operational business descriptions can refine sector assignment and reporting-item applicability. Numerical checks can also compare measure definitions, units, periods, and business boundaries before generating review records. As more reporting years become available, repeated company comparisons can form a longer panel under a recorded score version. Finally, the evidence-linking approach can be evaluated against international sustainability frameworks using separate reporting-item mappings and tests for each standard.

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A Supplementary Tables

The supplementary tables retain the complete records behind the company, sector, and year-wise summaries. Values come from the saved CSV exports. Component abbreviations follow Equation (4): C is Completeness, Sp is Specificity, Cl is Clarity, and G is SDG breadth. Components are percentages, and TS is the combined score. Pctl. denotes the sector percentile, calculated from companies with strictly lower scores.

A.1 Complete financial-company results

Table S1 lists all 94 companies in Financial Services (excl. Insurance). FY end is the date reconstructed from the latest report per CIN because the original score export omitted report identifiers. N counts retained scoring text segments. Source: `table_S1_financial_companies.csv`.

Table S1. Complete results for the 94 financial companies. Pctl. denotes the saved sector percentile; FY end is a reconstructed date.

Company	FY end	C	Sp	Cl	G	TS	Pctl.	N
REC Limited	2026-03-31	56.01	57.69	98.08	82.35	67.56	98.9	52
Mahindra & Mahindra Financial Services Limited	2026-03-31	67.83	49.15	98.31	29.41	64.48	97.9	59
YES BANK LIMITED	2026-03-31	60.45	46.67	92.00	70.59	63.64	96.8	75
KOTAK MAHINDRA BANK LIMITED	2026-03-31	61.21	59.42	95.65	17.65	63.21	95.7	69
ICICI Bank Limited	2026-03-31	56.95	57.14	95.92	29.41	62.05	94.7	49
PNB Housing Finance Limited	2026-03-31	84.01	25.86	93.10	17.65	61.75	93.6	58
KARUR VYSYA BANK LIMITED	2026-03-31	54.78	45.10	100.00	58.82	61.32	92.6	51
Punjab National Bank	2026-03-31	54.21	42.11	92.98	82.35	61.15	91.5	57
SBI Cards and Payment Services Limited	2026-03-31	61.49	33.78	93.24	76.47	61.02	90.4	74
IDFC FIRST Bank Limited	2026-03-31	56.86	48.98	95.92	35.29	60.15	89.4	49
Tata Capital Limited	2026-03-31	59.04	53.70	92.59	17.65	60.01	88.3	54
Axis Bank Limited	2026-03-31	68.87	38.89	97.22	11.76	59.84	87.2	72
Cholamandalam Financial Holdings Limited	2026-03-31	67.74	25.76	95.45	58.82	59.80	86.2	66
HDB Financial Services Limited	2026-03-31	66.13	25.00	97.50	52.94	58.75	85.1	80
State Bank of India ("SBI/the Bank/Bank")	2026-03-31	57.05	45.00	88.33	47.06	58.69	84.0	60
Home First Finance Company India Limited	2026-03-31	67.46	38.00	98.00	5.88	58.57	83.0	50
THE KARNATAKA BANK LIMITED	2025-03-31	63.67	41.18	96.08	11.76	58.21	81.9	51
IndusInd Bank	2026-03-31	68.78	28.81	94.92	29.41	58.08	80.9	59
L&T Finance Limited	2026-03-31	59.79	44.44	98.15	11.76	58.05	79.8	54
MUTHOOT FINANCE LIMITED	2026-03-31	67.83	33.33	98.15	5.88	57.35	78.7	54
THE FEDERAL BANK LIMITED	2026-03-31	69.82	28.57	95.71	11.76	56.82	77.7	70
Power Finance Corporation Limited (PFC)	2026-03-31	68.78	34.04	89.36	11.76	56.77	76.6	47
AU Small Finance Bank Limited	2026-03-31	58.09	43.40	96.23	11.76	56.68	75.5	53
IIFL Finance Limited	2026-03-31	59.32	47.06	88.24	11.76	56.67	74.5	51
Cholamandalam Investment and Finance Company Limited	2026-03-31	66.23	26.92	90.38	35.29	56.17	73.4	52
Bandhan Bank Limited	2026-03-31	55.63	44.44	94.44	11.76	55.65	72.3	54
Aadhar Housing Finance Limited	2026-03-31	56.01	46.15	90.38	11.76	55.50	71.3	52

Continued on next page

Table S1 continued

Company	FY end	*C*	*Sp*	*Cl*	*G*	TS	Pctl.	*N*
Housing & Urban Development Corporation Limited	2026-03-31	56.10	31.43	91.43	47.06	54.86	70.2	70
Union Bank of India	2026-03-31	64.81	25.00	93.18	23.53	54.41	69.1	44
Shriram Finance Limited (SFL)	2026-03-31	55.72	41.82	89.09	11.76	53.83	68.1	55
Manappuram Finance Limited	2026-03-31	54.02	38.00	98.00	11.76	53.78	67.0	50
360 ONE WAM LIMITED	2026-03-31	68.68	27.16	90.12	0.00	53.64	66.0	81
Aditya Birla Capital Limited	2026-03-31	53.93	39.02	95.12	11.76	53.48	64.9	41
CANARA BANK	2026-03-31	52.98	38.46	96.15	11.76	53.14	63.8	52
CITY UNION BANK LIMITED	2026-03-31	50.90	27.45	98.04	47.06	52.91	62.8	51
Aptus Value Housing Finance India Limited	2026-03-31	65.94	19.15	97.87	11.76	52.87	61.7	47
Aavas Financiers Limited	2025-03-31	61.78	32.61	82.61	11.76	52.19	60.6	46
INDIAN RAILWAY FINANCE CORPORATION LIMITED	2026-03-31	57.62	26.23	96.72	17.65	52.03	59.6	61
Westlife Foodworld Limited	2026-03-31	54.30	34.09	100.00	0.00	51.95	58.5	44
BANK OF INDIA	2026-03-31	53.64	30.95	100.00	11.76	51.92	57.4	42
Punjab & Sind Bank	2026-03-31	64.05	21.88	90.62	11.76	51.48	56.4	32
IDBI Bank Limited	2026-03-31	56.29	32.20	94.92	0.00	51.16	55.3	59
BANK OF BARODA	2026-03-31	55.82	18.33	96.67	35.29	50.69	54.3	60
POONAWALLA FINCORP LIMITED	2026-03-31	57.24	30.43	86.96	11.76	50.59	53.2	46
UCO BANK	2026-03-31	54.97	25.00	97.92	11.76	50.25	52.1	48
ANAND RATHI WEALTH LIMITED	2026-03-31	64.14	22.86	88.57	0.00	50.23	51.1	35
Jio Financial Services Limited	2026-03-31	56.10	32.69	82.69	11.76	49.96	50.0	52
SUNDARAM FINANCE LIMITED	2026-03-31	65.85	15.79	86.84	11.76	49.62	48.9	38
BANK OF MAHARASHTRA	2026-03-31	52.22	27.78	94.44	11.76	49.29	47.9	36
GIC Housing Finance Limited	2026-03-31	59.79	15.09	94.34	11.76	48.49	46.8	53
NORTHERN ARC CAPITAL LIMITED	2026-03-31	31.88	47.83	100.00	11.76	48.28	45.7	23
Can Fin Homes Limited	2026-03-31	36.33	43.14	94.12	17.65	48.06	44.7	51
RBL Bank Limited	2026-03-31	52.13	20.00	92.50	23.53	47.71	43.6	40
FIVE-STAR BUSINESS FINANCE LIMITED	2026-03-31	54.78	19.61	98.04	0.00	47.40	42.6	51
Satin Creditcare Network Limited	2026-03-31	37.37	38.78	95.92	11.76	46.94	41.5	49
INDIAN OVERSEAS BANK	2026-03-31	53.55	26.15	76.92	17.65	46.41	40.4	65
Indian Bank	2026-03-31	56.29	18.18	90.91	0.00	46.15	39.4	33
UTKARSH SMALL FINANCE BANK LIMITED	2026-03-31	34.53	43.59	87.18	11.76	45.50	38.3	39
Muthoot Microfin Limited	2026-03-31	35.76	29.09	94.55	35.29	45.47	37.2	55
Capri Global Capital Limited	2025-03-31	32.07	29.27	90.24	52.94	44.95	36.2	41
Tata Investment Corporation Limited	2026-03-31	59.41	15.22	76.09	11.76	44.72	35.1	46
The South Indian Bank Limited	2026-03-31	35.38	30.23	100.00	11.76	44.40	33.0	43
Central Bank of India	2026-03-31	51.18	17.86	92.86	0.00	44.40	33.0	28
Fusion Finance Limited	2026-03-31	36.61	32.69	96.15	0.00	43.68	31.9	52
Spandana Sphoorty Financial Limited	2026-03-31	33.49	45.71	80.00	0.00	43.11	30.9	35
CSB BANK LIMITED	2026-03-31	36.05	26.09	93.48	11.76	42.12	28.7	46
JSW HOLDINGS LIMITED	2026-03-31	66.89	2.44	73.17	0.00	42.12	28.7	41
TVS Holdings Limited	2026-03-31	54.78	10.00	86.00	0.00	42.11	27.7	50
Sammaan Capital Limited	2025-03-31	35.48	26.67	91.11	11.76	41.59	26.6	45
Bajaj Holdings & Investment Limited	2026-03-31	55.72	8.00	84.00	0.00	41.49	25.5	25
PNB Gilts Ltd.	2025-03-31	37.37	29.27	87.80	0.00	41.29	24.5	41

Continued on next page

Table S1 continued

Company	**FY end**	*C*	*Sp*	*Cl*	*G*	**TS**	**Pctl.**	*N*
SBFC Finance Limited	2026-03-31	31.22	26.19	95.24	11.76	40.57	23.4	42
Vakrangee Limited	2025-03-31	33.49	15.38	94.23	29.41	39.80	22.3	52
IFCI Limited	2025-03-31	33.30	19.61	86.27	29.41	39.40	21.3	51
DCB Bank Limited	2026-03-31	33.77	25.40	90.48	0.00	39.22	20.2	63
EQUITAS SMALL FINANCE BANK LIMITED	2026-03-31	34.15	18.18	87.88	0.00	36.69	19.1	33
INDIA SHELTER FINANCE CORPORATION LIMITED	2026-03-31	34.72	19.51	78.05	11.76	36.53	18.1	41
Raymond Limited	2026-03-31	31.98	21.43	78.57	11.76	36.11	17.0	42
SURYODAY SMALL FINANCE BANK LIMITED	2026-03-31	30.37	16.67	90.00	0.00	35.15	16.0	30
SUMMIT SECURITIES LIMITED	2026-03-31	24.98	13.51	94.59	11.76	34.14	14.9	37
Arman Financial Services Limited	2025-03-31	29.99	18.18	81.82	0.00	33.81	13.8	33
Ugro Capital Limited	2026-03-31	28.48	13.64	90.91	0.00	33.67	12.8	22
IndoStar Capital Finance Limited	2025-03-31	29.23	11.43	91.43	0.00	33.41	11.7	35
SG Finserve Limited	2026-03-31	29.52	15.79	84.21	0.00	33.39	10.6	19
Dolat Algotech Limited	2025-03-31	28.86	9.09	95.45	0.00	33.36	9.6	22
Religare Enterprises Limited	2025-03-31	38.98	6.00	78.00	0.00	32.99	8.5	50
MAS FINANCIAL SERVICES LIMITED	2026-03-31	35.29	10.20	77.55	0.00	32.69	7.4	49
Nahar Capital and Financial Services Limited	2025-03-31	27.53	10.81	89.19	0.00	32.09	6.4	37
Dhani Services Limited	2025-03-31	32.64	13.64	72.73	0.00	31.69	5.3	44
Capital India Finance Limited	2025-03-31	25.83	19.35	77.42	0.00	31.62	4.3	31
PILANI INVESTMENT AND INDUSTRIES CORPORATION LIMITED	2026-03-31	28.00	11.54	84.62	0.00	31.59	3.2	26
Fedbank Financial Services Limited	2025-03-31	31.98	12.12	75.76	0.00	31.58	2.1	33
Nalwa Sons Investments Limited	2025-03-31	31.13	10.26	74.36	0.00	30.40	1.1	39
Vardhman Holdings Limited	2025-03-31	35.00	2.78	75.00	0.00	29.83	0.0	36

A.2 Complete sector summary

Table S2 lists all 72 recorded sectors, with N giving the company count. Peer group Yes identifies sectors with at least five companies; No identifies the smaller groups pooled in Table 12. Median is the middle score, and SD is standard deviation. Source: `table_S2_all_sectors.csv`.

Table S2. All 72 sector groups, including the 25 small groups combined in the main sector table.

Sector	N	Mean	Median	Range	SD	Peer group
Chemicals Manufacturing	101	49.56	47.91	32.18–75.34	9.32	Yes
Financial Services (excl. Insurance)	94	48.17	49.79	29.83–67.56	9.88	Yes
Pharmaceuticals	65	51.18	50.03	32.38–73.72	10.06	Yes
Basic Metals	60	50.80	47.92	34.61–75.58	10.87	Yes
Electrical Equipment	51	48.80	47.68	26.88–67.03	9.56	Yes
Computer Programming and Information Technology Services	49	50.24	46.07	32.89–73.98	12.25	Yes
Non-Metallic Mineral Products	40	50.36	48.72	35.66–69.20	8.69	Yes
Food Products Manufacturing	38	47.57	44.90	30.04–69.86	9.60	Yes
Motor Vehicles	37	51.76	49.88	33.29–70.12	9.98	Yes
Machinery and Equipment	36	49.46	47.20	35.61–69.67	10.06	Yes
Rubber and Plastics	36	49.22	46.09	35.44–73.87	9.90	Yes
Auxiliary Financial Services	31	47.26	49.32	27.67–61.39	9.08	Yes
Electricity, Gas, Steam and AC Supply	31	53.67	55.43	31.95–74.23	11.49	Yes
Fabricated Metal Products	27	50.29	48.95	35.03–65.72	8.66	Yes
Textiles	26	47.44	47.53	31.77–67.65	10.16	Yes
Wholesale Trade	24	45.46	40.71	28.05–64.86	12.39	Yes
Civil Engineering	23	48.31	46.79	31.99–66.15	10.84	Yes
Retail Trade	21	43.06	41.42	34.22–56.45	6.12	Yes
Construction	20	48.73	47.94	36.58–63.25	8.32	Yes
Insurance and Pension Funding	20	51.52	53.52	29.92–64.17	10.12	Yes
Real Estate Activities	19	47.90	47.54	35.17–63.42	8.78	Yes
Human Health Activities	18	46.37	45.50	33.92–62.86	8.12	Yes
Information Service Activities	18	47.39	44.03	33.97–64.19	9.71	Yes
Computer, Electronic and Optical Products	17	47.46	46.26	32.93–65.45	7.59	Yes
Other Transport Equipment	17	50.83	48.79	37.67–66.89	9.39	Yes
Other Manufacturing	16	46.11	43.02	30.59–63.81	10.19	Yes
Telecommunications	15	50.63	46.05	36.57–69.19	11.43	Yes
Accommodation	14	48.67	46.75	31.28–67.57	10.06	Yes
Mining and Quarrying	13	47.40	41.94	29.01–73.16	13.27	Yes
Beverages Manufacturing	10	47.30	41.20	34.70–67.71	13.18	Yes
Other (Division 99)	10	48.39	43.36	34.80–73.44	12.12	Yes
Wholesale/Retail Trade - Motor Vehicles	10	49.25	45.00	37.40–67.92	11.47	Yes
Architecture and Engineering	9	48.50	43.34	37.49–64.86	10.06	Yes
Coke and Refined Petroleum	9	52.32	53.21	38.20–69.63	11.13	Yes
Paper Products	9	43.27	44.26	33.42–48.93	4.88	Yes
Broadcasting	8	43.65	46.96	31.66–52.74	7.64	Yes
Land Transport	7	47.42	47.52	34.72–63.63	10.11	Yes
Water Supply and Waste Management	7	43.33	38.97	36.73–56.76	8.29	Yes
Wood Products	7	46.18	46.66	36.16–64.51	9.72	Yes
Education	6	42.43	40.46	33.28–55.70	7.74	Yes
Furniture	6	51.28	51.48	36.17–65.29	9.25	Yes
Other Professional/Technical Activities	6	48.43	48.50	38.73–57.90	8.70	Yes
Scientific R&D	6	44.90	43.30	38.95–52.11	5.71	Yes
Warehousing and Transport Support	6	54.55	54.34	37.82–67.03	10.34	Yes
Wearing Apparel	6	42.33	41.82	37.17–48.78	4.50	Yes

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Table S2 continued

Sector	*N*	Mean	Median	Range	SD	Peer group
Leather Products	5	50.00	45.56	42.14–58.86	7.80	Yes
Water Transport	5	48.37	46.32	38.70–61.22	9.75	Yes
Food and Beverage Service	4	49.10	50.81	36.77–58.03	8.90	No
Motion Picture and Sound Production	4	37.26	37.29	31.12–43.36	5.54	No
Other (Division 34)	4	46.46	45.61	42.37–52.22	4.14	No
Publishing Activities	4	45.19	43.98	39.22–53.56	6.05	No
Tobacco Products	4	59.47	59.74	51.04–67.36	8.16	No
Travel Agency Activities	4	41.41	41.21	37.59–45.64	4.39	No
Employment Activities	3	42.67	38.73	33.77–55.52	11.40	No
Office Administrative Support	3	40.11	38.60	37.37–44.35	3.73	No
Printing	3	41.54	38.85	38.83–46.95	4.68	No
Agriculture, Forestry and Fishing	2	36.50	36.50	34.33–38.67	3.07	No
Air Transport	2	48.24	48.24	43.74–52.74	6.36	No
Rental and Leasing	2	37.52	37.52	34.21–40.82	4.67	No
Repair and Installation of Machinery	2	38.88	38.88	37.94–39.82	1.33	No
Specialized Construction	2	49.71	49.71	39.59–59.83	14.31	No
Sports and Recreation	2	44.44	44.44	43.24–45.63	1.69	No
Advertising and Market Research	1	41.36	41.36	41.36–41.36	0.00	No
Head Office and Management Consultancy	1	39.29	39.29	39.29–39.29	0.00	No
Other (Division 00)	1	31.58	31.58	31.58–31.58	0.00	No
Other (Division 02)	1	70.97	70.97	70.97–70.97	0.00	No
Other (Division 40)	1	37.80	37.80	37.80–37.80	0.00	No
Other (Division 67)	1	27.56	27.56	27.56–27.56	0.00	No
Other (Division 89)	1	45.77	45.77	45.77–45.77	0.00	No
Public Administration	1	42.42	42.42	42.42–42.42	0.00	No
Security and Investigation	1	47.25	47.25	47.25–47.25	0.00	No
Unclassified	1	41.54	41.54	41.54–41.54	0.00	No

A.3 Complete paired-year records

Table S3 preserves both report records and every score component for the eight paired-year companies in Table 13. Report ID identifies the database record. FY2024–25 and FY2025–26 end on 31 March 2025 and 31 March 2026, respectively. Source: `table_S3_paired_year_components.csv`.

Table S3. All 16 report records for the eight paired-year companies, with report identifiers and complete score components.

Company	Report ID	FY	*C*	*Sp*	*Cl*	*G*	TS
ESAF Small Finance Bank Ltd.	1207	2024–25	34.34	30.23	93.02	11.76	42.59
ESAF Small Finance Bank Ltd.	457	2025–26	54.59	41.30	93.48	11.76	54.10
G R Infraprojects Limited (GRIL)	1302	2024–25	33.87	19.57	82.61	0.00	35.94
G R Infraprojects Limited (GRIL)	723	2025–26	36.52	29.79	76.60	0.00	38.87
GOPAL SNACKS LIMITED	1186	2024–25	34.82	31.58	89.47	0.00	41.30
GOPAL SNACKS LIMITED	1396	2025–26	26.96	60.00	86.67	11.76	47.29
Praj Industries Ltd.	1239	2024–25	30.94	20.59	91.18	17.65	38.55
Praj Industries Ltd.	443	2025–26	55.72	25.00	93.75	17.65	50.30
SML MAHINDRA LIMITED	1085	2024–25	33.49	35.29	96.08	11.76	44.38
SML MAHINDRA LIMITED	751	2025–26	33.77	34.15	90.24	11.76	42.98
Sudarshan Chemical Industries Limited	1008	2024–25	75.02	36.00	96.00	29.41	62.95
Sudarshan Chemical Industries Limited	388	2025–26	73.60	39.22	94.12	52.94	65.32
Sudarshan Colorants India Limited	1007	2024–25	39.45	37.78	86.67	52.94	49.74
Sudarshan Colorants India Limited	385	2025–26	39.26	61.11	97.22	52.94	58.78
V.I.P INDUSTRIES LIMITED	1367	2024–25	29.42	48.15	92.59	11.76	45.91
V.I.P INDUSTRIES LIMITED	337	2025–26	30.18	53.85	80.77	11.76	45.56

A.4 Pharmaceutical-company examples

Table S4 presents the four highest and three lowest scores in the recorded Pharmaceuticals group. Selection identifies these score positions; the companies are not a random group. Source: `table_S4_pharmaceutical_examples.csv`.

Table S4. Seven pharmaceutical-company examples with all score components and saved sector percentiles.

Company	Selection	*C*	*Sp*	*Cl*	*G*	TS	Pctl.
Dabur India Limited	Top four	73.13	57.38	95.08	82.35	73.72	98.5
Lupin Limited	Top four	68.68	47.30	98.65	82.35	69.63	96.9
Dr. Reddy's Laboratories Limited	Top four	73.42	43.55	85.48	82.35	67.76	95.4
Ipca Laboratories Limited	Top four	85.62	28.36	91.04	52.94	66.26	93.8
Syncom Formulations (India) Limited	Bottom three	36.33	11.43	88.57	0.00	35.67	3.1
Jagsonpal Pharmaceuticals Limited	Bottom three	30.84	9.30	90.70	0.00	33.27	1.5
Kopran Limited	Bottom three	32.92	23.68	60.53	0.00	32.38	0.0